

Design of a Three-Phase N-Channel MOSFET Inverter and Analog Front-End for a 2.5–5 kW Battery-Powered BLDC/PMSM Drive with Field-Oriented Control

Hardware Design Report

Power stage: Infineon IPT015N10N5 (100 V, 1.5 m Ω) OptiMOS™ 5 MOSFET + Texas Instruments UCC27211A-Q1 half-bridge gate driver

Controller: STMicroelectronics STM32G473RCT6 Switching frequency: 40 kHz Control: field-oriented control (FOC) with space-vector PWM

Abstract

This report presents the component-level design of the power stage, the gate-drive stage, and the analog front-end of a three-phase voltage-source inverter for a brushless DC / permanent-magnet synchronous (BLDC/PMSM) motor under field-oriented control (FOC) with space-vector pulse-width modulation (SVPWM). The drive is battery-powered: a 24 V pack whose terminal voltage ranges from 19 V to 30 V with state of charge, delivering up to 2.5 kW; and a 48 V pack ranging from 39 V to 60 V, delivering up to 5 kW. The variation of the pack voltage is central to the design: the *highest* bus voltage (60 V) sets the blocking and dv/dt stresses, while the *lowest* bus voltage (19 V) produces the largest current for a given power and therefore the worst case for conduction loss and for the current-sense chain. Each switch position uses two Infineon IPT015N10N5 100 V OptiMOS™ 5 MOSFETs in parallel (four devices per half-bridge leg, twelve total), and each leg is driven by one Texas Instruments UCC27211A-Q1 120 V, 3.7 A/4.5 A half-bridge gate driver supplied at 12 V. The report (i) verifies the electrical compatibility of the selected devices across the full battery-voltage range; (ii) derives the gate-drive passive network — bootstrap capacitor, bootstrap-diode requirement, asymmetric gate resistors, gate–source pull-downs, and decoupling — from published design equations; and (iii) designs the three remaining analog blocks of the input stage: the in-line shunt phase-current-sense chain, the DC-bus and phase voltage-sense chains, and a do-not-populate RC snubber. The worst-case continuous phase current is derived as ≈ 119 A RMS (≈ 169 A peak) and used to size a 0.1 m Ω metal-element shunt read by a Texas Instruments INA240A2 current-sense amplifier with enhanced PWM rejection. Every quantitative claim and every design equation is supported by a primary source.

Index Terms

BLDC, PMSM, field-oriented control (FOC), space-vector PWM, MOSFET gate driver, bootstrap, asymmetric gate resistor, in-line current sensing, shunt resistor, INA240, voltage divider, STM32G4, RC snubber, single ground.

I. INTRODUCTION AND SCOPE

THE target system is a non-isolated three-phase voltage-source inverter, built on a single shared ground, intended to drive any BLDC/PMSM motor of roughly 0.5 to 5 kW under field-oriented control with space-vector PWM and Hall-effect rotor-position sensing. The controller is an STMicroelectronics STM32G473RCT6 — a motor-control microcontroller based on an Arm® Cortex®-M4 core that integrates up to five 12-bit analog-to-digital converters (ADCs), operational amplifiers usable as a programmable-gain amplifier (PGA) or unity-gain follower with their outputs routed internally to an ADC, analog comparators, advanced motor-control timers, and a voltage reference buffer (VREFBUF) [14]. The gate driver is the Texas Instruments UCC27211A-Q1 [3], and each switch is an Infineon OptiMOS™ 5 IPT015N10N5 (100 V, 1.5 m Ω) [1], [2]. The microcontroller, the gate driver, and the choice of power MOSFET are fixed inputs to this work.

The report addresses, in one place, both the power/gate-drive stage and the analog input stage:

- 1) the power switch and gate driver, their key parameters, and a compatibility assessment across the full battery-voltage range (Sections II–VI);

TABLE I
GIVEN DESIGN PARAMETERS.

Parameter	Value	Rationale / source
Power source	Battery (24 V or 48 V pack)	Bus voltage varies with state of charge
DC-bus voltage range	24 V pack: 19 V to 30 V 48 V pack: 39 V to 60 V	$V_{\text{bus}}^{\text{max}} = 60 \text{ V}$ (full 48 V pack); $V_{\text{bus}}^{\text{min}} = 19 \text{ V}$ (depleted 24 V pack)
Output power	$\approx 2.5 \text{ kW}$ (24 V pack) / $\approx 5 \text{ kW}$ (48 V pack)	Motor power scales with bus voltage
Switching frequency f_{sw}	40 kHz	Above the audible band ($\sim 20 \text{ kHz}$), so the PWM tone is inaudible
Control	FOC with SVPWM	Sinusoidal, rotor-field-oriented current control [12], [13]
Position sensing	Hall-effect sensors	Six 60° sectors per electrical revolution; FOC angle by interpolation
Power switch	IPT015N10N5, $\times 2$ in parallel per position	100 V, 1.5 m Ω OptiMOS TM 5, optimized for high-current applications [1], [2]
Gate driver	UCC27211A-Q1, $\times 1$ per leg	120 V boot, 3.7 A/4.5 A half-bridge driver with integrated boot diode [3], [4]
Gate-drive supply V_{DD}	12 V	Within the device operating range 8 V to 17 V [3]

- 2) the gate-drive passive network, derived quantitatively (Section VII);
- 3) the *phase-current measurement* chain (in-line shunt + amplifier + ADC interface) for all three phases (Section VIII);
- 4) the *voltage measurement* chains for the DC bus and for each of the three phase nodes (Section IX); and
- 5) an *RC snubber* across the power switches (Section X),

followed by the grounding/noise strategy, thermal considerations, and the bill of materials. For each analog block the report explains the operating principle, derives the governing equations, computes numerical values for the present specification, and selects a specific, orderable component.

Because the inverter accepts any 24 V or 48 V battery with no assumed chemistry, the design is dimensioned for the worst case of each electrical stress rather than for a single nominal operating point. The system uses a *single shared ground*; noise immunity is obtained through layout discipline rather than through a separate analog ground plane (Section XI).

II. DESIGN SPECIFICATION

Table I lists the given design parameters. The control method is field-oriented control with space-vector PWM throughout; six-step (trapezoidal) commutation is discussed only as historical background in Section III. Rotor position is obtained from Hall-effect sensors.

A. Worst-case phase current

The current-handling hardware must be dimensioned for the largest phase current the inverter can carry, since this sets both the conduction loss in the MOSFETs and the shunt/amplifier scaling. For a sinusoidally modulated three-phase inverter using SVPWM, the maximum fundamental phase-voltage amplitude is $\hat{V}_{\text{ph}} = V_{\text{bus}}/\sqrt{3}$, so the maximum RMS phase voltage is

$$V_{\text{ph,rms}} = \frac{V_{\text{bus}}}{\sqrt{6}}. \quad (1)$$

With three phases and displacement power factor $\text{PF} = \cos \varphi$, the AC power is $P = 3 V_{\text{ph,rms}} I_{\text{ph}} \text{PF}$. Taking converter efficiency $\eta \rightarrow 1$ (the conservative assumption that maximizes current) and solving for the RMS phase current,

$$I_{\text{ph}} = \frac{\sqrt{6}}{3} \frac{P}{V_{\text{bus}} \text{PF}} \approx 0.816 \frac{P}{V_{\text{bus}} \text{PF}} \quad (2)$$

which is largest at minimum bus voltage and minimum power factor. Table II evaluates (2) at the four supply/power corners ($\text{PF} = 0.9$).

TABLE II
OPERATING CORNERS AND RESULTING PHASE CURRENT FROM (2) (PF = 0.9).

Supply / motor	V_{bus} (V)	P (kW)	I_{ph} (A,rms)	$\hat{I}_{\text{ph}}$ (A,pk)
24 V pack, 2.5 kW	19 (min)	2.5	119.4	168.8
24 V pack, 2.5 kW	30 (max)	2.5	75.6	106.9
48 V pack, 5 kW	39 (min)	5.0	116.3	164.5
48 V pack, 5 kW	60 (max)	5.0	75.6	106.9

The limiting corner is 2.5 kW at 19 V: $I_{\text{ph}}^{\text{max}} \approx 119$ A RMS, $\hat{I}_{\text{ph}}^{\text{max}} \approx 169$ A peak. The corresponding worst-case DC-bus current is $P/V_{\text{bus}}^{\text{min}} = 2500/19 \approx 132$ A. Because two devices are paralleled in each switch position, each MOSFET carries approximately half of the phase current, i.e. ≈ 60 A RMS, which is far below the 2×250 A ($T_C = 100^\circ\text{C}$) continuous rating of the paralleled pair [1]. The per-device conduction loss at this corner is $P = I^2 R_{\text{DS(on)}} \approx 60^2 \times 1.5 \text{ m}\Omega \approx 5.4$ W (rising toward ~ 8 W as $R_{\text{DS(on)}}$ increases with junction temperature), which requires adequate cooling (Section XII). All current-sense quantities below refer to this limiting corner.

III. BACKGROUND AND THEORY OF OPERATION

A. Why an inverter is required to drive a BLDC/PMSM machine

Despite its name, the brushless DC motor is a synchronous AC machine: it has a permanent-magnet rotor and a three-phase stator winding, and it lacks the mechanical commutator and brushes that a conventional DC motor uses to switch current between windings as the rotor turns [12]. The commutation function once performed mechanically by the brushes must therefore be performed electronically. The inverter is the circuit that does this: it converts the fixed-polarity DC bus voltage (here supplied by a battery) into the set of controlled, alternating phase currents needed to keep the stator magnetic field rotating ahead of the rotor and thereby produce continuous torque [12].

This design uses *field-oriented control* (FOC) with *space-vector PWM* (SVPWM). In FOC the measured three-phase currents are transformed (Clarke and Park transformations) into a rotor-aligned reference frame, where a direct-axis (d) and a quadrature-axis (q) current are regulated independently; the q -axis current produces torque while the d -axis current is normally held at zero [12]. The resulting voltage demand is synthesized by SVPWM, which selects and time-weights the inverter's switching states to approximate the commanded voltage vector with the best use of the available bus voltage [13]. Compared with the simpler six-step (trapezoidal, 120°) scheme — in which the six switches step through six discrete field orientations per electrical revolution — FOC drives sinusoidal phase currents and produces smooth torque with low ripple, which is why it is adopted here. The rotor angle that the Park transformation and SVPWM require is obtained from the Hall-effect sensors, interpolated between the six 60° sector transitions to recover a continuous electrical angle.

B. How the inverter circuit works

The power stage is a three-phase bridge: three identical half-bridge “legs,” one per motor phase, each consisting of a high-side switch and a low-side switch in series across the DC bus, with the motor phase connected to the midpoint. In this design each switch is two IPT015N10N5 MOSFETs in parallel, so one leg is four MOSFETs and the full bridge is twelve.

Within a leg, the two switches operate complementarily — never both on at once, which would short-circuit the DC bus (“shoot-through”). Turning on the high-side switch connects the phase to the positive bus; turning on the low-side switch connects it to ground; turning both off lets the phase float. Modulating the three legs with SVPWM circulates sinusoidal currents through the phases in the sequence that produces a rotating stator field [13]. The MOSFET body diodes (together with the motor inductance) provide the freewheeling path for the inductive phase current during the dead-time between switch states; the IPT015N10N5 body diode is characterized for this purpose ($V_{\text{SD}} \approx 0.85$ V, $Q_{\text{rr}} \approx 316$ nC) [1]. The resistive on-state of the MOSFET ($R_{\text{DS(on)}}$) as low as $1.5 \text{ m}\Omega$, again halved by paralleling) keeps the conduction loss low at currents on the order of 100 A [1], [5].

TABLE III
KEY PARAMETERS OF THE IPT015N10N5 [1].

Symbol	Parameter	Value / note
V_{DS}	Drain-source voltage	100 V (absolute maximum)
$R_{DS(on)}$	On-resistance at $V_{GS} = 10$ V	1.3 m Ω typ. / 1.5 m Ω max.
I_D	Continuous drain current	353 A ($T_C = 25^\circ\text{C}$); 250 A ($T_C = 100^\circ\text{C}$)
$V_{GS(th)}$	Gate threshold voltage	2.2 / 3.0 / 3.8 V (min/typ/max)
$V_{plateau}$	Miller-plateau voltage	4.4 V typ. (at $I_D = 100$ A)
C_{iss}	Input capacitance	12 nF typ. / 16 nF max.
C_{oss}	Output capacitance	1.8 nF typ.
C_{rss}	Reverse-transfer capacitance	80 pF typ. / 140 pF max.
$R_{G,i}$	Internal gate resistance	1.4 Ω typ. / 2.1 Ω max.
Q_g	Total gate charge (0–10 V)	169 nC typ. / 211 nC max.
Q_{gs}	Gate-source charge	53 nC
Q_{gd}	Gate-drain (Miller) charge	34 nC typ. / 51 nC max.
g_{fs}	Transconductance	140 S min. / 280 S typ.
V_{SD}	Body-diode forward voltage	0.85 V typ. / 1.2 V max.
Q_{rr}	Body-diode reverse-recovery charge	≈ 316 nC typ.
$V_{GS,max}$	Gate-source voltage limit	± 20 V

C. Purpose of each component

- **Power MOSFETs (IPT015N10N5 $\times 2$ per position):** the controlled switches that connect each phase to the buses. Two in parallel halve the effective on-resistance and share the conduction loss and heat; the positive temperature coefficient of $R_{DS(on)}$ makes the sharing self-stabilizing [5].
- **Gate driver (UCC27211A-Q1):** converts the controller's PWM logic into the high-current 12 V gate drive needed to switch the MOSFETs quickly, level-shifts the high-side signal to the floating switch node, and enforces under-voltage lockout (UVLO) [3]. A gate driver is required because a logic output (often 3.3 V, ≤ 1 A) can neither fully enhance a power MOSFET nor supply the peak current for fast switching, and because the high-side gate must be referenced to a node that swings to the bus voltage [3], [5].
- **Bootstrap capacitor:** the floating energy reservoir that supplies the high-side driver while the switch node is high (Section VII-A).
- **Bootstrap diode (internal):** a one-way valve that refills the bootstrap capacitor from V_{DD} when the switch node is low and blocks the bus voltage when it is high [3], [7].
- **Gate resistors:** set and limit the gate current, shape dv/dt for EMI, and damp/balance the paralleled gates [5].
- **Gate-source pull-down resistors:** hold the MOSFETs off at power-up and add dv/dt immunity against false turn-on [5].
- **Decoupling capacitors:** supply the fast gate-current pulses locally so that the 12 V rail and the driver bias do not collapse during switching [3].
- **DC-link capacitance:** stiffens the bus against the large half-bridge ripple current and limits the commutation overshoot within the 100 V device rating.

IV. POWER SWITCH: INFINEON IPT015N10N5

The IPT015N10N5 is an N-channel OptiMOSTM 5 power transistor in a TO-Leadless (PG-HSOF-8, "TOLL") package, which Infineon describes as suited to high-frequency switching and intended for light-electric-vehicle, forklift, and power-tool drives [1], [2]. The parameters relevant to this design are summarized in Table III; all values are from the datasheet (Rev. 2.4) unless noted [1].

A property that directly justifies paralleling is the positive temperature coefficient of $R_{DS(on)}$ (about +0.7 to +1 %/K in power MOSFETs): a device that carries more current heats up, its resistance rises, and current is pushed

TABLE IV
KEY PARAMETERS OF THE UCC27211A-Q1 [3].

Symbol	Parameter	Value
V_{DD}	Supply (operating / abs. max)	8 V to 17 V / 20 V
$I_{O,source} / I_{O,sink}$	Peak output current	3.7 A / 4.5 A
$V_{DD,UVLO}$	UVLO threshold on V_{DD} (rising)	7.0 V (hysteresis 0.5 V)
$V_{HB,UVLO}$	UVLO threshold on high side (HB–HS, rising)	6.7 V (hysteresis 1.1 V)
V_{HB}	Max boot-supply voltage	120 V (HB–HS $\leq$ 20 V)
V_{HS}	Switch-node range	$-(24 - V_{DD})$ V (i.e. -12 V at $V_{DD} = 12$ V) to 105 V recommended
V_F	Boot-diode forward drop	0.85 V (@100 μ A) / 1.05 V (@100 mA) max.
R_D	Boot-diode dynamic resistance	0.55 Ω typ.
t_{PD}	Propagation delay	20 ns typ.
t_R/t_F	Rise / fall time (1 nF)	7.2 ns / 5.5 ns
I_{HB}	Boot quiescent current	0.12 mA max.
SR_{HS}	Max high-side slew rate	50 V/ns
C_{HB}	Recommended boot capacitor	0.022 μ F to 0.1 μ F (gate-charge dependent)
C_{VDD}	Recommended V_{DD} decoupling	0.22 μ F to 4.7 μ F

back toward the cooler device, so paralleled MOSFETs self-balance their currents [5]. This is the standard reason MOSFETs (unlike bipolar transistors) are well suited to parallel operation [5].

V. GATE DRIVER: TEXAS INSTRUMENTS UCC27211A-Q1

The UCC27211A-Q1 is an AEC-Q100-qualified 120 V high-side/low-side half-bridge gate driver that drives two N-channel MOSFETs from independent inputs; its floating high-side stage can operate to 120 V, and a 120 V bootstrap diode is integrated on-chip, so no external boot diode is mandatory [3], [4]. The relevant parameters, all from datasheet SLUSCG0C, are listed in Table IV [3].

TI divides the device into five functional blocks — input stages, UVLO protection, level shift, boot diode, and output stages [3]. The inputs are independent HI/LI with ≈ 68 k Ω pull-downs and TTL-compatible thresholds with wide hysteresis for noise immunity; they tolerate -10 V to $+20$ V and are independent of V_{DD} , so they accept both microcontroller logic and analog-controller signals [3]. Separate monitors on V_{DD} and on the floating V_{HB} – V_{HS} supply hold the outputs low until the bias is valid. The level shifter translates the ground-referenced high-side command up to the floating high-side stage with tight (≈ 4 ns) channel-to-channel delay matching. Both output stages provide 3.7 A source / 4.5 A sink with fast, low-impedance edges and a 20 ns propagation delay [3].

The largest output current is required *around the Miller-plateau voltage* (~ 5 V), because it is there that the gate driver must push the Miller charge Q_{gd} through the plateau in the intended transition time; the datasheet states the peak-current requirement as $I_{PEAK} = Q_{GD}/dt$, and the 3.7 A capability is checked against this in Section VII-D [3], [5].

VI. COMPATIBILITY ASSESSMENT

Table V checks the device pair against the full battery-voltage range.

The pair is electrically sound across the full battery range. At the highest bus voltage (60 V, full 48 V pack) the 100 V class gives a $\approx 1.67\times$ DC margin and about 40 V of reserve for the $L di/dt$ turn-off overshoot of V_{DS} ; this makes control of the turn-off speed (via the asymmetric gate network of Section VII-E) more important than it would be on a fixed 48 V bus. The only point that genuinely required quantitative verification — because the design parallels two high- Q_g MOSFETs per switch — is whether one channel of the UCC27211A-Q1 can charge and discharge the *doubled* gate charge fast enough; this is treated in Section VII-D and in the bootstrap sizing of Section VII-B, and is the reason the standard 0.1 μ F bootstrap capacitor must be revisited for this load.

TABLE V
COMPONENT COMPATIBILITY CHECK.

Check	Requirement	This design	Outcome
Blocking voltage	$V_{DS} \gg V_{bus}^{\max}$ incl. ringing	100 V vs 60 V; $\approx 1.67\times$ DC margin	Pass — ≈ 40 V headroom for turn-off overshoot [1]
High-side supply voltage	$V_{HB} > V_{bus}^{\max}$	120 V boot, 105 V HS rec., vs 60 V	Pass [3]
Gate-drive level	$V_{GS,\max} > V_{DD} \geq$ full enhancement	± 20 V limit, driven at 12 V	Pass [1], [3]
UVLO	$V_{DD} > \text{UVLO}$	12 V vs 7 V (V_{DD}), 6.7 V (HB)	Pass [3]
Negative node ringing	HS tolerates undershoot	-12 V transient capability at $V_{DD} = 12$ V	Pass [3]
Current capability	$I_D \gg$ leg current	2×250 A (100 °C) vs ≈ 60 A/device worst case	Pass [1]
Drive strength vs charge	Sufficient current through the Miller plateau for the <i>paralleled</i> charge	see Section VII-D	Pass

VII. GATE-DRIVE PASSIVE NETWORK DESIGN

Throughout, the gate charge of one *switch position* is twice that of a single device, because two MOSFETs are paralleled:

$$Q_{g,\text{pos}} = 2Q_g = 2 \times 211 \text{ nC} = 422 \text{ nC (worst case, to 10 V)} [1]. \quad (3)$$

Because the gate is driven to 12 V rather than the 10 V at which Q_g is specified, and because the charge curve is nearly flat above ~ 7 V, a $\approx 10\%$ margin is added, giving a design value of $Q_{g,\text{pos}} \approx 465 \text{ nC}$ per switch position.

A. Bootstrap operation and the Miller plateau

a) *Why an N-channel-only half-bridge needs a bootstrap supply:* N-channel MOSFETs are preferred for the half-bridge because, for a given die area and voltage, they have far lower $R_{DS(\text{on})}$ than P-channel devices. But an N-channel MOSFET turns on only when its gate is several volts above its source. For the low-side device the source is at ground, so a ground-referenced 12 V drive works directly. For the high-side device the source is the switch node, which rises to the full bus voltage when that device is on; to keep it on, the gate must be driven ≈ 12 V above the bus — a voltage that exists nowhere in a single-supply system [3], [5]. The bootstrap circuit creates this floating high-side supply cheaply, using only a diode and a capacitor [7].

b) *How it works:* When the low-side switch is on, the switch node (and the HS pin) is pulled to ground; the bootstrap capacitor, connected between HB and HS, charges from the 12 V V_{DD} rail through the bootstrap diode to $\approx V_{DD} - V_F \approx 11$ V [3], [7]. When the high-side then turns on, the switch node rises to the bus voltage; the bootstrap diode reverse-biases (blocking the bus), and the charged capacitor “rides up” on the switch node, holding HB ≈ 11 V above HS and thus supplying the floating bias that keeps the high-side MOSFET fully enhanced [3], [7]. The capacitor is refilled each cycle when the low-side next turns on — which is why a minimum low-side on-time (refresh) is required, and why Section VII-B verifies both the droop and the recharge [7], [8].

c) *The Miller plateau:* During each turn-on the gate voltage rises, pauses on a flat “Miller plateau” (≈ 4.4 V here [1]), then rises again. The pause occurs because, once the channel carries the full load current, all the gate-drive current is diverted into discharging the gate–drain capacitance C_{GD} so that the drain voltage can swing — during this interval V_{GS} is clamped and only the drain voltage moves [5]. The plateau interval is exactly the window in which the device simultaneously sustains high voltage and high current, so it is where the switching loss is generated [5]. The Miller charge Q_{gd} that must be pushed through the plateau is doubled by paralleling ($\approx 102 \text{ nC}$ worst case), and the *speed* of the voltage transition is set by how much current the driver can force into the gate at the plateau level — not by its peak current at full V_{DD} [5]. This is why Balogh emphasizes that the most important attribute of a gate driver is its source/sink capability *around the plateau* [5].

B. Bootstrap capacitor

The bootstrap capacitor C_{BOOT} must supply the entire charge needed to turn on the high-side device and to feed the high-side quiescent current during the high-side on-time, while drooping by no more than the allowed ΔV_{BS} [7], [8]. The governing relations are

$$\Delta V_{BS} \leq V_{DD} - V_F - V_{GS,\min} - V_{DS,\text{on}}, \quad (4)$$

$$Q_{\text{TOTAL}} = Q_{g,\text{pos}} + \frac{I_{QBS} + I_{LK}}{f_{\text{sw}}} + Q_{LS}, \quad C_{BOOT} \geq \frac{Q_{\text{TOTAL}}}{\Delta V_{BS}}, \quad (5)$$

with an additional $\sim 20\%$ margin recommended for tolerances and for capacitance loss when the diode is integrated [8].

a) *Numbers:* The boot-diode drop at the relevant charging current is $V_F \approx 0.9\text{ V}$ [3]; requiring the high-side gate to stay fully enhanced, $V_{GS,\min} \approx 10\text{ V}$, with low-side $V_{DS,\text{on}} \approx 0.1\text{ V}$, the *allowed* droop from (4) is $\Delta V_{BS} \leq 12 - 0.9 - 10 - 0.1 = 1.0\text{ V}$. A conservative design target of $\Delta V_{BS} = 0.5\text{ V}$ is chosen, which keeps the floating supply at $\approx 10.6\text{ V}$, well above the 6.7 V high-side UVLO [3]. The quiescent/leakage charge over the worst-case high-side on-time is negligible: with $I_{QBS} = 0.12\text{ mA}$ [3], a high duty $D_{\text{max}} \approx 0.95$, and $t_{HS} = 0.95/40\text{ kHz} = 23.75\text{ }\mu\text{s}$,

$$Q_{\text{leak}} \approx (0.12\text{ mA})(23.75\text{ }\mu\text{s}) \approx 2.9\text{ nC} \ll Q_{g,\text{pos}}. \quad (6)$$

The gate charge dominates, so $Q_{\text{TOTAL}} \approx 465\text{ nC} + 3\text{ nC} \approx 468\text{ nC}$ and

$$C_{BOOT} \geq \frac{468\text{ nC}}{0.5\text{ V}} \approx 0.94\text{ }\mu\text{F}. \quad (7)$$

Applying the recommended margin and allowing for DC-bias capacitance loss in Class-II ceramics, the selected value is $C_{BOOT} = 2.2\text{ }\mu\text{F}$, X7R, $\geq 25\text{ V}$, placed directly between the HB and HS pins [3], [8].

b) *Why the datasheet's $0.022\text{ }\mu\text{F}$ to $0.1\text{ }\mu\text{F}$ range does not apply here:* That range is correct for a *single* MOSFET with $\sim 30\text{--}90\text{ nC}$ of gate charge [3]; the paralleled pair here requires about $5\times$ more charge, which is exactly why the datasheet states the value “is dependant on the gate charge of the high-side MOSFET” [3]. Using $0.1\text{ }\mu\text{F}$ would give a droop of $468\text{ nC}/0.1\text{ }\mu\text{F} \approx 4.7\text{ V}$ per cycle — enough to under-drive the device or even trip the high-side UVLO. The larger $2.2\text{ }\mu\text{F}$ value is therefore not optional for this load.

c) *Recharge check:* The capacitor replaces only the $\approx 468\text{ nC}$ removed each cycle. At the worst duty the low-side on-time is $\approx 1.25\text{ }\mu\text{s}$; the required average charging current is $468\text{ nC}/1.25\text{ }\mu\text{s} \approx 0.37\text{ A}$, while the integrated $0.55\text{ }\Omega$ boot diode can initially supply $(12 - 0.9)/0.55 \approx 20\text{ A}$ [3]. Recharge is therefore comfortable at 40 kHz . At very low motor speed with high-side duty near 100% , the refresh window shrinks and a periodic low-side refresh pulse may be required — a known limitation of all bootstrap supplies [7], [8].

C. Bootstrap diode

The UCC27211A-Q1 *integrates* the bootstrap diode (anode to V_{DD} , cathode to HB); it is rated 120 V , has a typical dynamic resistance of $0.55\text{ }\Omega$, a forward drop of $\approx 0.9\text{--}1.05\text{ V}$ at 100 mA , and a 20 ns turn-off time [3]. For this design the internal diode is adequate: Section VII-B shows that both the charging current and the droop are comfortably within its capability, and its 120 V rating exceeds the 60 V bus by a wide margin [3]. An optional external diode in parallel may be added to further reduce the forward drop and per-cycle droop; if used, it must block $\geq V_{\text{bus}} + V_{DD}$ (a 100 V part is chosen to match the MOSFET class), recover quickly ($t_{rr} \leq 50\text{ ns}$) or be a Schottky, and carry the peak charge current ($\geq 1\text{ A}$) [7], [8]. Because the function is already on-chip, this component is an upgrade, not a necessity.

D. Gate resistors and peak-current / Miller check

a) *Damping requirement:* The external gate resistor for critically damping the gate-loop resonance formed by the source inductance L_S and C_{iss} is [5]

$$R_{\text{GATE,OPT}} = 2\sqrt{\frac{L_S}{C_{iss}}} - (R_{DRV} + R_{G,i}). \quad (8)$$

For the low-inductance TOLL package (L_S on the order of 1 nH) and $C_{iss} = 12$ nF, $2\sqrt{L_S/C_{iss}} \approx 0.58 \Omega$, which is already exceeded by the driver output impedance plus the internal gate resistance ($R_{DRV} + R_{G,i}$). The result is negative, meaning the loop is *already over-damped by the driver impedance and the internal 1.4 Ω gate resistance alone* [5], [1]. The external gate resistor is therefore chosen not for damping but to (i) keep the shared driver output current within rating, (ii) shape dv/dt for EMI, and (iii) balance the two paralleled devices.

b) *Individual resistors*: Best practice for paralleled MOSFETs is an *individual* gate resistor at each gate (two per switch position) rather than a single shared resistor, so that any tendency of the two devices to oscillate against one another is damped and their switching is balanced. As a symmetric baseline, $R_{G,ext} = 2.2 \Omega$ (E24) per MOSFET is close to the 1.8Ω at which the IPT015N10N5 switching times are characterized [1]; Section VII-E refines this into the asymmetric network that is actually adopted.

c) *Peak-current / dv/dt check*: With the gate held at the plateau $V_{plateau} = 4.4$ V [1], the network would demand, per device,

$$I_G = \frac{V_{DD} - V_{plateau}}{R_{G,ext} + R_{G,i}} = \frac{12 - 4.4}{2.2 + 1.4} \approx 2.1 \text{ A}, \quad (9)$$

i.e. ≈ 4.2 A for the pair, which exceeds the 3.7 A source limit; the driver therefore limits the current to 3.7 A and itself sets the transition speed. The time to deliver the combined Miller charge (2×51 nC = 102 nC max) at 3.7 A is

$$t_{\text{Miller}} = \frac{Q_{gd,pos}}{I_{O,source}} = \frac{102 \text{ nC}}{3.7 \text{ A}} \approx 28 \text{ ns}, \quad (10)$$

which, at the worst-case 60 V bus, gives

$$\frac{dV_{DS}}{dt} \approx \frac{60 \text{ V}}{28 \text{ ns}} \approx 2.1 \text{ V/ns}, \quad (11)$$

which is moderate (good for conducted EMI) and far below the 50 V/ns HS slew-rate limit [3]. This confirms the key compatibility question: *even with the doubled gate charge of paralleling, one channel of the UCC27211A-Q1 switches the pair in under 30 ns* — precisely the use case for the device's 3.7 A/4.5 A rating [3].

E. Asymmetric (split) gate-resistor network

A single resistor forces one value to serve both switching edges, which is a compromise. This is replaced by an *asymmetric* network that sets the turn-on and turn-off speeds independently — slower, softer turn-on and fast, low-impedance turn-off — which is the configuration adopted in the final design.

a) *Topology*: Per MOSFET, an always-present series resistor R_{on} carries the turn-on (source) current, and a branch of R_{off} in series with an OFF Schottky diode D_{off} is placed in parallel with it. D_{off} is oriented anode-to-gate, cathode-to-driver-output, so it conducts *only when the gate is pulled down* (turn-off / dv/dt clamp). Hence

$$R_{g(on)} = R_{on}, \quad R_{g(off)} = R_{on} \parallel (R_{off} + R_D), \quad (12)$$

so at turn-on the gate sees only R_{on} , and at turn-off it sees the lower parallel combination [5], [9].

b) *Design direction — slow on, fast off*: For a hard-switched half-bridge the preferred asymmetry is slower turn-on, faster turn-off [9], [10], [11]. Faster turn-on raises the di/dt of the drain current, which increases the reverse-recovery current of the complementary device's freewheeling body diode; the IPT015N10N5 body diode stores $Q_{rr} \approx 316$ nC, so deliberately softened turn-on limits that recovery peak, its associated loss, and the EMI it generates [1], [9], [10]. Turn-off, conversely, benefits from being fast and low-impedance: it reduces turn-off loss and, crucially, holds the gate of the *off* device firmly at ground so that the Miller current injected through C_{GD} when the complementary switch slews the node cannot raise V_{GS} to the threshold and cause self (false) turn-on [10], [11]. The 100 V device on a bus up to 60 V has enough voltage reserve (≈ 40 V) to absorb the slightly higher turn-off overshoot that fast turn-off produces; because that reserve is smaller than on a fixed 48 V bus, the ability to control the turn-off edge independently is all the more valuable. The asymmetry also reinforces the driver's own 4.5 A-sink / 3.7 A-source imbalance [3].

c) *Value selection:*

- $R_{on} = 3.3\ \Omega$ (per MOSFET). Raised above the $2.2\ \Omega$ baseline to soften the turn-on edge. The resulting per-device plateau current is $(12 - 4.4)/(R_{on} + R_{G,i}) = 7.6/(3.3 + 1.4) = 1.62\ \text{A}$ ($\approx 3.2\ \text{A}$ for the pair) — now set by the resistor rather than the $3.7\ \text{A}$ driver limit — giving a controlled dv/dt of $60/(102\ \text{n}/3.2) \approx 1.9\ \text{V/ns}$ worst case [1], [3].
- $R_{off} = 1.0\ \Omega$ (per MOSFET). With the OFF-diode dynamic resistance $R_D \approx 0.1\text{--}0.15\ \Omega$ (from the PMEG4030ER conduction characteristic, $V_F \approx 0.46\text{--}0.54\ \text{V}$) [9], $R_{g(\text{off})} = R_{on} \parallel (R_{off} + R_D) = 3.3 \parallel 1.15 \approx 0.85\ \Omega$, an effective on/off resistance ratio of about 3.9:1.

The internal gate resistance $R_{G,i} = 1.4\ \Omega$ is in series with the die and cannot be bypassed [1], so the total turn-off gate-loop resistance is $R_{G,i} + R_{g(\text{off})} = 2.25\ \Omega$; reducing R_{off} below $\sim 1\ \Omega$ yields diminishing returns, so $1.0\ \Omega$ is well placed. At this value the per-device turn-off plateau current is $4.4/2.25 \approx 1.96\ \text{A}$ ($\approx 3.9\ \text{A}$ for the pair, near the $4.5\ \text{A}$ sink limit), giving a turn-off transition of $\approx 26\ \text{ns}$ versus $\approx 32\ \text{ns}$ at turn-on — the desired asymmetry [3].

d) *OFF diode selection:* D_{off} must block the reverse voltage that appears across R_{on} at turn-on ($\approx I_{on,\text{pk}} R_{on} \approx 1.62 \times 3.3 \approx 5.3\ \text{V}$ plus ringing), carry the turn-off current pulse ($\approx 2\ \text{A}$ peak per gate), recover instantly, and have low dynamic resistance. A Schottky satisfies all four. The selected component is the **Nexperia PMEG4030ER** — $40\ \text{V}$, $3\ \text{A}$ Schottky rectifier with low V_F ($\approx 0.46\text{--}0.54\ \text{V}$), $I_{FSM} = 50\ \text{A}$, SOD-123W [9]. Its $40\ \text{V}$ rating gives $> 7\times$ margin over the $\sim 5\ \text{V}$ reverse stress, and its low V_F and small junction capacitance keep the turn-off path clean [9].

e) *Resistor power and package:* The average power in a gate resistor is a fraction of the total gate-drive loss $P_G = Q_g V_{DD} f_{sw}$ [5]. Per device, with $Q_g \approx 232\ \text{nC}$ at $12\ \text{V}$, $P_G \approx 0.11\ \text{W}$, split between turn-on and turn-off; the external resistor sees only its fraction of the gate-loop resistance, so $P_{R_{on}} \approx 26\ \text{mW}$ and $P_{R_{off}} \approx 6\ \text{mW}$ — both far below $1/16\ \text{W}$. The resistor does see short current pulses ($\sim 1.6\text{--}2\ \text{A}$), but the per-transition energy in R_{on} is $\approx 0.6\ \mu\text{J}$, negligible against any chip resistor's pulse rating. A **0805 thick-film, 1/8 W** package is chosen for both R_{on} and R_{off} ($> 4\times$ average-power margin and good pulse robustness) [5].

F. Gate-source pull-down resistors

A gate-to-source resistor holds each MOSFET definitely off before the driver outputs become active at power-up, and adds dv/dt protection against Miller-induced turn-on [5]. In normal operation the driver's low output impedance and $4.5\ \text{A}$ sink dominate this protection; the pull-down's main job is the power-up state. A standard $R_{GS} = 10\ \text{k}\Omega$ per MOSFET is large enough not to load the driver and small enough to hold the gate down at power-up. A capacitive-divider estimate confirms the margin: when the complementary switch imposes the worst-case $60\ \text{V}$ on the off device, the worst-case gate rise through the divider $C_{rss}/C_{iss} = 80\ \text{pF}/12\ \text{nF} \approx 0.0067$ would be only $\approx 0.40\ \text{V}$ even with a *floating* gate — still far below the $2.2\ \text{V}$ threshold — and the active $4.5\ \text{A}$ sink holds the real value lower still [1], [3]. A **0603, 1/10 W** package suffices (the resistor dissipates $12^2/10\ \text{k}\Omega = 14.4\ \text{mW}$ when the gate is high).

G. Decoupling capacitors

a) *V_{DD} (low-side) decoupling:* Whenever the LO output sources gate current, an equal pulse is drawn from V_{DD} , so a low-ESR ceramic must sit directly between V_{DD} and V_{SS} [3]. Sized for a tolerable ripple ΔV with the same charge argument as the bootstrap, $C_{VDD} \geq Q_{g,\text{pos}}/\Delta V = 465\ \text{nC}/0.5\ \text{V} \approx 0.93\ \mu\text{F}$. TI recommends $0.22\ \mu\text{F}$ to $4.7\ \mu\text{F}$ [3]; the chosen value is $C_{VDD} = 4.7\ \mu\text{F}\ \text{X7R} + 0.1\ \mu\text{F}$ per driver, one set for each of the three drivers.

b) *HB (high-side) decoupling:* This role is performed by the bootstrap capacitor itself (Section VII-B), placed directly between HB and HS; the separate $0.022\ \mu\text{F}$ to $0.1\ \mu\text{F}$ recommendation is only the minimum HF decoupling and is exceeded here by the computed $2.2\ \mu\text{F}$ [3].

c) *12 V bulk:* A shared bulk reservoir (e.g. $22\ \mu\text{F}$ to $47\ \mu\text{F}$) on the auxiliary $12\ \text{V}$ rail feeds the three drivers and absorbs the aggregate gate-drive pulses [3].

d) *DC-link decoupling:* Because the source is a battery connected through a cable harness, the source impedance (internal battery resistance plus harness inductance) is significant, so local DC-link decoupling is even more important than on a stiff laboratory bus. The power stage carries a ripple current on the order of $100\ \text{A}$ (up to $\approx 132\ \text{A}$ worst case at the depleted $24\ \text{V}$ pack) and must be decoupled by a low-ESR/ESL bank (film and/or

TABLE VI
GATE-DRIVE PASSIVE BILL OF MATERIALS (PER LEG / PER BOARD).

Ref.	Function	Value	Per leg	3 legs
C_{BOOT}	Bootstrap capacitor (HB-HS)	$2.2\ \mu\text{F}$ X7R $\geq 25\ \text{V}$	1	3
C_{VDD}	V_{DD} decoupling	$4.7\ \mu\text{F}$ X7R + $0.1\ \mu\text{F}$	1 set	3 sets
R_{on}	Turn-on resistor (per MOSFET)	$3.3\ \Omega$, 0805, 1/8 W	4	12
R_{off}	Turn-off resistor (per MOSFET)	$1.0\ \Omega$, 0805, 1/8 W	4	12
D_{off}	OFF Schottky diode	Nexperia PMEG4030ER (40 V, 3 A)	4	12
R_{GS}	Gate-source pull-down	$10\ \text{k}\Omega$, 0603, 1/10 W	4	12
D_{BOOT} (opt.)	External parallel boot diode	100 V fast/Schottky $\geq 1\ \text{A}$	0–1	0–3

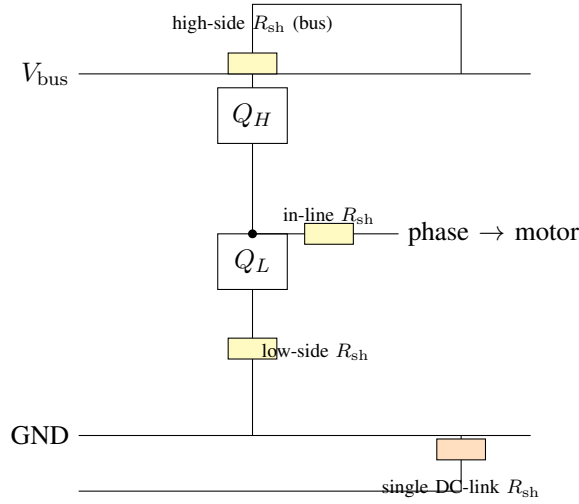


Fig. 1. Possible shunt positions in one inverter leg: high-side (in series with the leg from the bus), low-side (in series with the leg to ground), in-line (in series with the phase conductor), and the single DC-link shunt in the common return. The Hall/magnetic alternative replaces the shunt with a field sensor around the phase conductor.

ceramic) close to the half-bridge, sized for the RMS ripple current and to keep the commutation overshoot within the 100 V rating. Since the highest battery voltage is 60 V, the bus capacitors should be rated $\geq 80\ \text{V}$ (film) or $\geq 100\ \text{V}$ (ceramic, allowing for DC-bias derating). Precise sizing depends on the harness inductance and is a board-level task.

Plus the active devices: $12 \times$ **IPT015N10N5** (four per leg) and $3 \times$ **UCC27211A-Q1**.

VIII. PHASE-CURRENT SENSING

A. Survey and comparison of current-sense topologies

Four families of inverter current sensing exist; their measurement node is shown in Fig. 1 for one half-bridge, and they are compared in Table VII.

For field-oriented control the inner current loop needs the *actual* phase current continuously and independently of duty cycle. The low-side and high-side shunts measure current only while their switch conducts, becoming blind near 100 % and 0 % duty respectively; the single DC-link shunt requires duty-dependent reconstruction. The *in-line* (in-phase) shunt, placed in series with the phase conductor, observes the true phase current at any duty cycle, which is why it is chosen here, in line with TI's TIDA-00913 reference design [16]. Its cost is a sense element whose common-mode potential swings from rail to rail at the switching frequency with high dv/dt , which dictates the amplifier requirements in Section VIII-C.

TABLE VII
COMPARISON OF CURRENT-SENSE TOPOLOGIES FOR A THREE-PHASE INVERTER.

Attribute	Low-side shunt	High-side shunt	In-line (in-phase) shunt	Magnetic / Hall
Common-mode voltage at sense element	Near ground (≈ 0)	At/near V_{bus}	Swings rail-to-rail at f_{sw} with high dv/dt	Galvanically isolated; no CM
Measures true phase current at all duty cycles	No — only while Q_L conducts; blind near 100 % duty	No — only while Q_H conducts	Yes — continuous, at any duty	Yes
Amplifier requirement	Low-CM CSA	High-CM CSA	High-CM CSA with PWM rejection	— (sensor is the amplifier)
DC capability	Yes	Yes	Yes	Yes (Hall) / No (CT, Rogowski)
Suitability for FOC	Limited (reconstruction needed)	Limited	Direct (per-phase, continuous)	Good but costlier/larger

B. Shunt selection

A metal-element (metal-strip) shunt is used for its low inductance and low temperature coefficient. The shunt voltage is $v_{\text{sh}}(t) = R_{\text{sh}} i(t) + \text{ESL } di/dt$; at 40 kHz with fast switching edges the inductive term must be negligible against the 0.1 m Ω resistive term, which a metal-strip element with < 2 nH achieves. At the worst-case continuous current (≈ 119 A RMS) the shunt dissipates $I^2 R_{\text{sh}} \approx 119^2 \times 0.1 \text{ m}\Omega \approx 1.4$ W; a 0.1 m Ω , 5 W element gives ample margin. An alternative is two extremely common 0.2 m Ω , 2512 elements in parallel ($= 0.1$ m Ω , each dissipating ≈ 0.7 W).

C. Current-sense amplifier and gain

The amplifier must (i) tolerate a common-mode voltage that swings from slightly below ground (body-diode conduction during dead-time, ≈ -1 V) to $V_{\text{bus}} = 60$ V plus ringing overshoot; (ii) reject the large common-mode dv/dt at each switching edge so the output does not ring; (iii) be *bidirectional*, since the phase current is AC; and (iv) have a bandwidth well above 40 kHz. The Texas Instruments INA240 satisfies all four: it is a voltage-output, zero-drift current-sense amplifier with a -4 to $+80$ V common-mode range *independent of supply*, with enhanced PWM rejection explicitly designed to suppress the common-mode PWM transients of motor drives, offered in four fixed gains (20, 50, 100, 200 V/V) [15]. The -4 V lower limit accepts the sub-ground excursion, and the $+80$ V upper limit leaves 20 V of reserve above the 60 V bus for ringing. The PWM-rejection benefit is application-dependent rather than a single specified frequency; TI's validated TIDA-00913 reference design demonstrates accurate in-line operation up to PWM switching frequencies of 100 kHz, so the 40 kHz SVPWM here is comfortably within the validated range [15], [16].

On a 3.3 V analog supply, the bidirectional output at zero current is centered by the two reference pins of the INA240, whose internal divider sets the output midpoint at $V_{\text{ref}} = (V_{\text{REF1}} + V_{\text{REF2}})/2$ [15]; tying REF1 to ground and REF2 to the 3.3 V rail gives $V_{\text{ref}} = 1.65$ V. The output is then

$$V_{\text{out}} = V_{\text{ref}} + G R_{\text{sh}} i_{\text{ph}}. \quad (13)$$

With a rail-to-rail output the usable single-sided swing is ≈ 1.45 V (0.2 V margin to each rail). Choosing gain $G = 50$ (INA240A2), the linear full-scale peak current is

$$\hat{I}_{\text{FS}} = \frac{1.45}{G R_{\text{sh}}} = \frac{1.45}{50 \times 0.1 \text{ m}\Omega} = 290 \text{ A peak (205 A RMS)}. \quad (14)$$

The worst-case peak of 169 A produces an output excursion of $G R_{\text{sh}} \hat{I}_{\text{ph}} = 50 \times 0.1 \text{ m}\Omega \times 169 = 0.85$ V, i.e. $V_{\text{out}} \in [0.80, 2.50]$ V, leaving ≈ 0.5 V before clipping. This reserve is deliberately kept for transient overshoot and

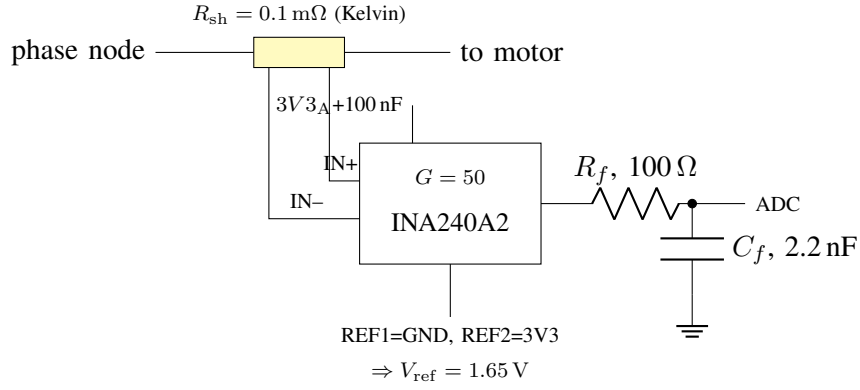


Fig. 2. In-line per-phase current-sense chain: Kelvin-connected $0.1 \text{ m}\Omega$ metal-strip shunt, INA240A2 ($G = 50$) with mid-scale reference, and a light 723 kHz RC into the STM32G4 ADC. Three identical chains are used (phases U, V, W).

lets an over-current (OC) threshold be set at $\approx 220 \text{ A}$ peak ($\approx 1.3 \times \hat{I}_{\text{ph}}^{\text{max}}$), detectable by an STM32G4 analog comparator or in software before the amplifier saturates. The quantization and offset, referred to current, are

$$\Delta I_{\text{LSB}} = \frac{V_{\text{REF+}}/2^N}{G R_{\text{sh}}} = \frac{3.3/4096}{50 \times 0.1 \text{ m}\Omega} = 0.16 \text{ A/LSB}, \quad (15)$$

$$I_{\text{os}} = \frac{V_{\text{os(max)}}}{R_{\text{sh}}} = \frac{25 \mu\text{V}}{0.1 \text{ m}\Omega} = 0.25 \text{ A}, \quad (16)$$

both negligible against a 119 A signal (the INA240 zero-drift architecture gives $V_{\text{os}} \leq 25 \mu\text{V}$) [15]. The gain-50 / $0.1 \text{ m}\Omega$ pair is therefore preferred over a gain-20 / $0.25 \text{ m}\Omega$ pair, which would dissipate $\approx 4.6 \text{ W}$ per shunt for the same full scale.

D. ADC interface and output filtering

For FOC the phase current is sampled *synchronously* with the PWM carrier (typically at the counter peak/valley) so that the sampling instant coincides with the middle of the switching state, where the inductor ripple is at its average value; this naturally rejects the 40 kHz ripple and its harmonics without a heavy filter [18]. A heavy anti-alias filter would instead add phase lag to the very signal on which the current loop closes, degrading the loop. The CSA output therefore uses only a light first-order RC filter ($R_f = 100 \Omega$, $C_f = 2.2 \text{ nF}$) that (i) band-limits the high-frequency switching noise and ringing (corner $f_c = 1/2\pi R_f C_f = 723 \text{ kHz}$, far above the kHz-range loop content but damping the tens-of-MHz ringing) and (ii) acts as a charge reservoir for the ADC sample-and-hold. The low output impedance of the INA240 drives this network and the ADC sampling capacitor with ample settling margin. STM32G4 hardware oversampling/averaging may be enabled for further noise reduction at the cost of bandwidth [18]. Fig. 2 shows the complete per-phase chain.

IX. VOLTAGE SENSING

A. Requirements

Hardware voltage measurement is needed for (i) the DC-bus voltage — for over-/under-voltage protection, modulation-index scaling, and power calculation — and (ii) each of the three phase-node voltages — for output monitoring, phase-loss / motor-disconnect detection, and diagnostics. The phase-voltage chains also leave a hardware path open for future back-EMF-based position observation without a board change, although the present design commutates from Hall sensors and does not rely on phase-voltage feedback for control. All four channels share the inverter ground (the design is non-isolated and uses a single shared ground), so each is a grounded resistive divider feeding an STM32G4 ADC channel. The bus channel is slow (protection/scaling); the phase channels preserve more bandwidth for monitoring and for the optional back-EMF capability.

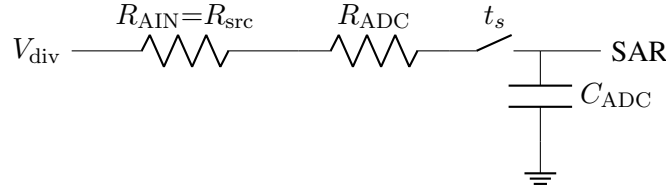


Fig. 3. First-order model of the STM32G4 ADC input stage used in (20): the divider source impedance R_{src} in series with the internal switch resistance charges the sampling capacitor C_{ADC} over the sampling window t_s .

B. Resistive-divider design

The ADC conversion range is 0 to $V_{\text{REF}+}$; using $V_{\text{REF}+} = V_{\text{DDA}} = 3.3 \text{ V}$ maximizes the range. The divider ratio $k = (R_{\text{top}} + R_{\text{bot}})/R_{\text{bot}}$ maps the maximum measured voltage to near full scale with reserve. Choosing $R_{\text{top}} = 110 \text{ k}\Omega$ and $R_{\text{bot}} = 4.7 \text{ k}\Omega$,

$$k = \frac{R_{\text{top}} + R_{\text{bot}}}{R_{\text{bot}}} = \frac{110 + 4.7}{4.7} = 24.4, \quad V_{\text{ADC}} = \frac{V_{\text{in}}}{k}. \quad (17)$$

This maps $V_{\text{bus}} = 60 \text{ V} \rightarrow 2.46 \text{ V}$ and $30 \text{ V} \rightarrow 1.23 \text{ V}$, with full scale at $\approx 73 \text{ V}$ — enough headroom to read transients above the 60 V nominal maximum. The same ratio is used for the phase dividers, since a phase node can reach the bus voltage. The standing current and dissipation at 60 V are

$$I_{\text{div}} = \frac{60}{R_{\text{top}} + R_{\text{bot}}} = 523 \mu\text{A}, \quad P_{R_{\text{top}}} = I_{\text{div}}^2 R_{\text{top}} = 30 \text{ mW}, \quad (18)$$

so all four dividers together draw $\approx 126 \text{ mW}$ — negligible. The top resistor sees the full bus voltage; a single thick-film 110 k Ω chip rated $\geq 150 \text{ V}$ (e.g. 0805/1206) is adequate at 60 V, but splitting R_{top} into two series 55 k Ω parts is recommended to halve the per-component voltage stress and improve ratio matching. The Thévenin source impedance presented to the ADC is

$$R_{\text{src}} = R_{\text{top}} \parallel R_{\text{bot}} = \frac{110\text{k} \times 4.7\text{k}}{110\text{k} + 4.7\text{k}} = 4.51 \text{ k}\Omega, \quad (19)$$

which is the key quantity for the buffer question of Section IX-C.

C. Does the ADC need an op-amp buffer? A quantitative answer

An STM32G4 ADC channel samples by connecting the source, through an internal multiplexer/switch resistance R_{ADC} , to a sampling capacitor C_{ADC} for the programmed sampling time t_s (Fig. 3). The capacitor must settle to within $\frac{1}{2}$ LSB of the source voltage. For an N -bit conversion this requires [20], [18]

$$t_s \geq (R_{\text{AIN}} + R_{\text{ADC}}) C_{\text{ADC}} \ln(2^{N+1}), \quad (20)$$

where $R_{\text{AIN}} = R_{\text{src}}$ is the external source impedance. ST does not publish $R_{\text{ADC}}/C_{\text{ADC}}$ directly for the G4; instead the datasheet gives a “maximum R_{AIN} versus sampling time” table and recommends choosing the sampling time from it for the given source impedance [18]. Using the AN2834 settling model with representative values $C_{\text{ADC}} \approx 5 \text{ pF}$, $R_{\text{ADC}} \approx 2 \text{ k}\Omega$, and $\ln(2^{13}) = 9.01$ for the 12-bit mode,

$$t_s \geq (4510 + 2000)(5 \times 10^{-12})(9.01) = 293 \text{ ns}. \quad (21)$$

At a 60 MHz ADC clock this is 17.6 cycles, whereas the STM32G4 offers sampling-time settings up to 640.5 cycles; the 47.5-cycle setting alone is 792 ns — more than $2.7\times$ the requirement. Because the bus and phase voltages need only kHz-rate refresh, spending a longer sampling time costs nothing. Therefore:

No external op-amp buffer is required for the voltage dividers.

The divider source impedance settles to 12-bit accuracy well within a trivially available sampling time; a small filter capacitor (Section IX-D) further reduces the effective source impedance at the sampling instant.

If a buffer were ever desired — for example to share one divider among several fast multiplexed channels, or to add an active filter — it should be realized with the *MCU’s own resources* rather than an external IC: the STM32G473

TABLE VIII
OP-AMP BUFFER BEFORE THE ADC: BENEFITS VERSUS COSTS.

Arguments for a buffer	Arguments against a buffer
Low output impedance $\Rightarrow$ fast settling, short sampling time, high multiplexed scan rate	Adds its own offset, drift, noise, and finite-bandwidth error to an <i>absolute</i> measurement
Isolates the divider from ADC charge-injection “kickback” and inter-channel crosstalk	Extra BOM cost, board area, and supply decoupling
Enables an active anti-alias filter	Becomes an additional calibration burden
Prevents resistive loading of the divider	Unnecessary here: the required t_s is small, (20)

TABLE IX
PHASE LOW-PASS FILTER DELAY $\phi = \arctan(f_e/f_c)$ (COMPENSATED IN FIRMWARE IF BACK-EMF OBSERVATION IS USED).

f_c	$f_e = 200$ Hz	$f_e = 500$ Hz	$f_e = 1000$ Hz
2 kHz	5.7°	14.0°	26.6°
3 kHz	3.8°	9.5°	18.4°

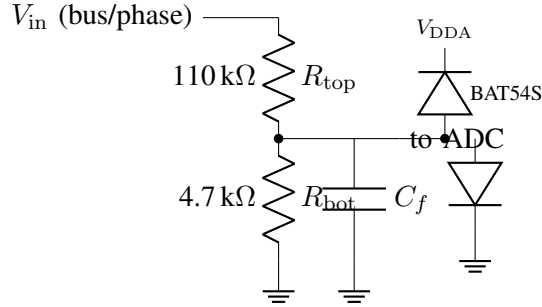


Fig. 4. Measurement divider (ratio 24.4), first-order filter capacitor, and Schottky clamp to V_{DDA} /GND. Identical networks serve the DC bus ($f_e \approx 1.6$ kHz) and the three phase nodes ($f_e \approx 2$ –3 kHz).

contains operational amplifiers that can be configured as unity-gain followers (or PGA) with their output routed *internally* to an ADC channel, which both removes the external-pin noise-pickup path and costs nothing on the BOM [19], [14]. ST documents exactly this follower-to-ADC use and recommends a 200 ns sampling time when reading an internal op-amp output [18], [19]. For the present low-rate voltage channels the conclusion is to omit the buffer and use a longer sampling time plus a filter capacitor.

D. Bus/phase filtering and protection

A capacitor C_f in parallel with R_{bot} forms a first-order low-pass filter with corner $f_c = 1/(2\pi R_{src} C_f)$. **Bus channel:** $C_f = 22$ nF gives $f_c = 1.6$ kHz, removing the switching ripple from the slow bus measurement. **Phase channels:** the filter must reject the 40 kHz PWM while passing the fundamental of the monitored phase voltage; a corner of $f_c \approx 2$ –3 kHz attenuates the carrier at the cost of a phase lag $\phi = \arctan(f_e/f_c)$, where f_e is the electrical frequency. If the phase voltages are used for back-EMF observation (the provisioned future capability), this lag is deterministic and is compensated in firmware (the observer shifts the sampled phase by ϕ); Table IX lists representative values. **Protection:** during dead-time/body-diode conduction a phase node swings to ≈ -1 V, scaled by the divider to $-1/24.4 = -41$ mV at the ADC pin — small, but a Schottky clamp (e.g. BAT54S) from the ADC node to ground and to V_{DDA} is added as insurance against over-/under-range transients.

X. RC SNUBBER ON THE SWITCHING NODE

A. Origin of the ringing

At each turn-off, the parasitic inductance L_p of the commutation loop (package leads, PCB traces, bus-capacitor ESL) resonates with the MOSFET output capacitance C_o (C_{oss} plus mounting capacitance), producing a damped oscillation on the switching node at

$$f_{\text{ring}} = \frac{1}{2\pi\sqrt{L_p C_o}}, \quad (22)$$

typically in the tens of MHz. The ringing stresses the device voltage, radiates EMI, and couples into signal lines. An RC damping snubber placed across the device adds a resistor R_s (in series with a capacitor C_s) that dissipates the resonant energy [21].

B. Design methodology

The rigorous, layout-independent procedure measures the ringing on the assembled board [21]:

- 1) Measure the ring period without a snubber, $T_1 = 1/f_{\text{ring}}$, on the switching node.
- 2) Add a known capacitor C_{test} across the device and measure the new period T_2 .
- 3) Since $T_1 = 2\pi\sqrt{L_p C_o}$ and $T_2 = 2\pi\sqrt{L_p(C_o + C_{\text{test}})}$, squaring and subtracting eliminates L_p :

$$L_p = \frac{T_2^2 - T_1^2}{4\pi^2 C_{\text{test}}}, \quad C_o = \frac{T_1^2}{4\pi^2 L_p}. \quad (23)$$

The damping resistor is set to the characteristic impedance of the resonance,

$$R_s = \sqrt{\frac{L_p}{C_o}}, \quad (24)$$

giving approximately critical damping [21]. The snubber capacitor is chosen a few times larger than C_o so that R_s dominates the loop, yet small enough to limit dissipation; the resistor power is the energy stored in C_s , charged and discharged once per switching cycle,

$$P_{R_s} = C_s V_{\text{bus}}^2 f_{\text{sw}}. \quad (25)$$

C. Estimated values (to be validated on the bench)

No assembled board exists yet, so L_p cannot be measured; the values below are a first estimate for footprint selection only and *must* be confirmed with the two-measurement method of (23) on the prototype. From the IPT015N10N5 datasheet, $C_{oss} = 1800$ pF (at the datasheet test condition; C_{oss} is strongly voltage-dependent) [1], [2]; adding ≈ 200 pF of mounting capacitance gives $C_o \approx 2$ nF. Taking a representative $L_p \approx 10$ nH for a compact layout,

$$f_{\text{ring}} = \frac{1}{2\pi\sqrt{(10\text{n})(2\text{n})}} = 35.6 \text{ MHz}, \quad (26)$$

$$R_s = \sqrt{\frac{10 \text{ nH}}{2 \text{ nF}}} = 2.24 \Omega \rightarrow 2.2 \Omega \text{ (E12)}. \quad (27)$$

Choosing $C_s = 2.2$ nF ($\approx C_o$), the worst-case dissipation (60 V bus) from (25) is

$$P_{R_s} = 2.2 \text{ nF} \times 60^2 \times 40 \text{ kHz} = 0.32 \text{ W per snubber}, \quad (28)$$

i.e. ≤ 1.9 W for all six switches if every footprint is populated. A ≥ 1 W resistor is specified for margin.

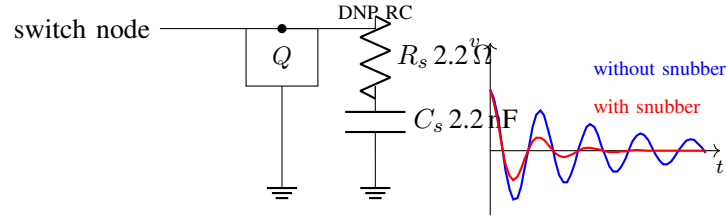


Fig. 5. DNP RC snubber across each switch and its qualitative effect on switching-node ringing: the damping resistor $R_s = \sqrt{L_p/C_o}$ removes the resonant energy. The values shown are first estimates to be replaced by bench measurements via (23).

D. Component selection

C_s must be C0G/NP0: Class-2 (X7R) ceramics lose much of their capacitance under DC bias and self-heat under the high-frequency ripple, whereas C0G is lossless and stable. A 2.2 nF, 100 V C0G part is the baseline (a 200 V part adds reserve against ringing overshoot). R_s must be non-inductive: an SMD thin/thick-film chip (1206 or 2512), never wirewound, so that its own inductance does not defeat the snubber [21]. The RC is placed directly across the drain–source of each switch with the shortest possible loop. Per the specification, all six RC sites are laid out as **do-not-populate (DNP)** footprints: if the prototype shows acceptable ringing they remain empty, and if not, they are populated using the bench-measured values from (23)–(24). Fig. 5 shows the placement and qualitative effect.

XI. GROUNDING, NOISE, AND FILTERING STRATEGY

The system uses a *single shared ground*: the high-current power return, the analog/signal reference, and all external interfaces sit on one common ground net. There is no separate analog ground plane; at 40 kHz and currents above 100 A, noise immunity is obtained through layout discipline rather than ground partitioning. Filtering is applied at each point where a sensitive analog signal is generated or where switching noise is injected:

- **CSA output** (Section VIII-D): a light 723 kHz RC (100 Ω/2.2 nF) per phase, relying on synchronous sampling rather than a delay-inducing anti-alias filter [18].
- **CSA input**: the input series resistors must be *matched* between IN+ and IN– to preserve CMRR; the INA240 enhanced PWM rejection is the primary defense against the common-mode transient, so a heavy input filter is deliberately avoided [15].
- **Bus divider**: 1.6 kHz RC (Section IX-D).
- **Phase dividers**: 2–3 kHz RC with firmware phase-delay compensation where used for back-EMF observation, plus Schottky clamps.
- **Analog supply**: a ferrite bead between V_{DD} and V_{DDA} with 1 μF + 100 nF decoupling on V_{DDA} ; separate 1 μF + 100 nF on V_{REF+} , plus the VREFBUF decoupling specified by ST if the internal reference is used [14].
- **INA240 supply**: 100 nF + 1 μF at the pin.
- **DC-link decoupling**: low-ESL ceramics in parallel with the bulk electrolytics, close to each half-bridge, to minimize the commutation-loop inductance L_p — attacking the *source* of the ringing the snubber would otherwise have to damp [21].
- **Layout**: Kelvin (four-wire) sensing across each shunt; short, tightly coupled differential sense traces routed away from the high- di/dt switching-current return paths; a compact commutation loop; and a solid, low-impedance ground pour so that the shared-path voltage drops are minimized. These layout measures are as important as the component filters at this power level.

XII. THERMAL CONSIDERATIONS

The MOSFETs are mounted to a liquid-cooled cold plate. At the worst-case corner each device carries ≈ 60 A RMS, giving a conduction loss of $\approx 60^2 \times 1.5 \text{ m}\Omega \approx 5.4 \text{ W}$ cold, rising toward $\sim 8 \text{ W}$ as $R_{DS(on)}$ increases with junction temperature; switching loss at 40 kHz and the shunt loss ($\approx 1.4 \text{ W}$ per phase at 119 A RMS) add to the budget. Using the device junction-to-case path (0.4 K/W) and an adequate thermal-interface material to the cold

TABLE X
ANALOG FRONT-END BILL OF MATERIALS. STOCK VARIES; CONFIRM AT ORDER TIME.

Function	Qty	Component / value	Key rating	Note
Current-sense amplifier	3	TI INA240A2PWR	$G=50$, -4 to 80 V CM	one per phase [22]
Phase shunt	3	$0.1\text{ m}\Omega$ metal element, 5 W	$\leq 2\text{ nH}$ ESL, Kelvin	confirm stock; see alt. below
(shunt alternative)	6	$0.2\text{ m}\Omega$ 2512 $\times 2$	2 W each	very common; $= 0.1\text{ m}\Omega$
CSA output filter R	3	$100\text{ }\Omega$ 0603 1%	—	with C_f
CSA output filter C	3	2.2 nF C0G 0603	50 V	f_c 723 kHz
Divider top	4(8)	$110\text{ k}\Omega$ ($2 \times 55\text{ k}\Omega$) 0805 1%	$\geq 150\text{ V}$	bus + 3 phases
Divider bottom	4	$4.7\text{ k}\Omega$ 0603 1%	—	ratio 24.4
Bus filter cap	1	22 nF C0G/X7R 0603	50 V	f_c 1.6 kHz
Phase filter cap	3	22 nF 0603	50 V	$f_c \approx 2\text{--}3\text{ kHz}$
ADC clamp diode	4	BAT54S dual Schottky	30 V	to V_{DDA}/GND
Snubber R (DNP)	6	$2.2\text{ }\Omega$ 1206 non-ind., $\geq 1\text{ W}$	—	footprint only
Snubber C (DNP)	6	2.2 nF C0G 1206	100 V (200 V opt.)	footprint only
Ferrite bead on VDDA	1	$600\text{ }\Omega@100\text{ MHz}$ 0805	—	$V_{DD} \rightarrow V_{DDA}$
Decoupling set	—	$100\text{ nF}/1\text{ }\mu\text{F}$ 0402/0603	$16\text{ V}+$	CSA, VDDA, VREF
Power MOSFET (ref.)	12	Infineon IPT015N10N5	$100\text{ V}/1.5\text{ m}\Omega$	power stage [22]

plate keeps the junction temperature within rating. The coolant inlet temperature and flow rate govern the junction temperature far more than the air ambient and are set per deployment. Low-inductance layout and adequate copper for the $\sim 169\text{ A}$ peak phase current and $\sim 132\text{ A}$ bus current remain board-level tasks, essential to realizing the calculated switching performance.

XIII. BILL OF MATERIALS (ANALOG FRONT-END)

Table X lists the analog front-end components. The active devices (INA240A2PWR, IPT015N10N5) and standard chip passives are high-volume parts; the only component to confirm for availability at order time is the high-power shunt, for which an alternative of two very common $0.2\text{ m}\Omega$ 2512 elements in parallel ($= 0.1\text{ m}\Omega$) is given. Quantities assume one three-phase board; the snubber components are DNP (footprints only).

XIV. CONCLUSION

A complete power, gate-drive, and analog input stage has been designed and costed for a 19 V to 60 V , $2.5\text{--}5\text{ kW}$, 40 kHz BLDC/PMSM FOC inverter. The Infineon IPT015N10N5 and Texas Instruments UCC27211A-Q1 form a compatible pair across the full battery range: at the highest bus voltage (60 V) the 100 V MOSFET gives a $\approx 1.67 \times$ DC blocking margin with about 40 V of reserve for turn-off overshoot, and at the lowest bus voltage (19 V) the worst-case current reaches $\approx 132\text{ A}$ on the bus ($\approx 60\text{ A}$ per device), far below the $2 \times 250\text{ A}$ rating of the paralleled pair. One channel of the 120 V , $3.7\text{ A}/4.5\text{ A}$ driver switches the *paralleled* gate charge in under 30 ns at a moderate $\approx 2\text{ V/ns}$, far below its 50 V/ns limit. The main consequence of paralleling two high- Q_g devices is that the bootstrap and V_{DD} decoupling capacitors must be $\approx 1\text{--}2\text{ }\mu\text{F}$ rather than the datasheet's nominal $0.1\text{ }\mu\text{F}$, a result that follows directly from the charge-droop equations. The complete gate-drive passive set is $C_{BOOT} = 2.2\text{ }\mu\text{F}$, $C_{VDD} = 4.7\text{ }\mu\text{F} + 0.1\text{ }\mu\text{F}$, an asymmetric gate network of $R_{on} = 3.3\text{ }\Omega$ with a parallel $R_{off} = 1.0\text{ }\Omega$ and OFF Schottky (Nexperia PMEG4030ER) per device for soft turn-on and fast turn-off, and $R_{GS} = 10\text{ k}\Omega$ per device.

The worst-case continuous phase current was derived from the supply/power range as ≈ 119 A RMS (≈ 169 A peak) at the 2.5 kW/19 V corner, and used to size a Kelvin 0.1 m Ω /5 W metal-element shunt read by an INA240A2 ($G=50$) with mid-scale reference, giving a ± 290 A-peak linear range at 0.16 A/LSB resolution and < 0.25 A offset. The in-line topology is justified against the low-side, high-side, single-shunt, and Hall alternatives by its continuous, duty-independent observability, consistent with TI TIDA-00913. For the voltage dividers (ratio 24.4) it was shown *quantitatively* that no external buffer is required: the 4.5 k Ω source settles to 12-bit accuracy in < 300 ns, well within the available sampling window, and the six STM32G473 internal op-amps remain available as a no-cost fallback. The RC snubber was sized from the published C_{oss} and the standard two-measurement ringing-extraction method and laid out as a DNP footprint on all six switches. Filtering was specified at the CSA output, the dividers, the analog supply/reference, and the DC bus, together with the single-shared-ground and Kelvin-sensing layout measures that are decisive at this power level. The next step is prototype bring-up: measuring the actual switching-node ringing to finalize the snubber, verifying the current-loop noise under synchronous sampling, and confirming shunt availability at BOM release.

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